Ruibo Wang

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EDUCATION

Delft University of Technology — *B.S. in Computer Science and Engineering* 09/2025 – **Now**

GPA 8.43/10, equivalent to 4.0/4.0 on the WES U.S. scale

Hebei University of Technology — *B.S. in Computer Science and Technology (GPA 3.83/4.0)* 08/2023 – 07/2024

PUBLICATIONS

- **Wang, R.**, Liu, F., Tiulpin, A., Sabuncu, M. R., Kim, H. "RaFL: An Empirical Study of Random Filter Losses for Image Reconstruction." International Conference on Learning Representations (ICLR) 2027. (Under Review)
- **Wang, R.**, Peng, F. "EquiFlow: Few-Step 3D MRI Reconstruction with an Equivariant MeanFlow Field." International Conference on Learning Representations (ICLR) 2027. (Under Review)
- **Wang, R.**, Shen, Z., Wu, H., Liang, D., Shang, K. "FSANet: Frequency-Spatial Aware Network for Image Segmentation." IEEE Transactions on Image Processing. (Under Review, arXiv:2609.16773)
- Shen, Z.*, **Wang, R.***, Jia, S., Yang, J., Liu, F., Shang, K. "ProFA-GAN: Hierarchical Prompt-Guided Frequency-Aware Generative Adversarial Network for 3D Brain Image Synthesis." Pattern Recognition and Computer Vision (PRCV) 2026. (Accepted)
- Wang, J.*, **Wang, R.***, Duruisseaux, V., Daftari, R., Gan, W., Jatyani, A. S., Bonev, B., Anandkumar, A. "Coarse-to-Fine 3D MRI Reconstruction via Resolution-Agnostic Neural Operators." Conference on Neural Information Processing Systems (NeurIPS) 2026. (Accepted, Poster)
- **Wang, R.**, Yang, J., Shen, Z., Chen, Z., Wang, C., Liang, D., Shang, K. "Scattering-Aware Implicit Neural Representation for CT Metal Artifact Reduction." IEEE Signal Processing Letters 2026. (Accepted)
- Hu, X.*, **Wang, R.***, Shen, Z.*, Zhang, L., Zeng, C., Liang, X., Song, J. "PanoFuse: Panoramic-Guided Sparse-View Dental CBCT Reconstruction." IEEE Transactions on Medical Imaging. (Under Review)
- **Wang, R.***, Luo, D.*, Shen, Z.*, Lin, Z.*, Li, H., Liu, F., Xu, N., Wang, J., Tao, T., Wu, X., Tian, W., Huang, C., Yu, B., Gao, J., Shen, C., Shang, K. "Multimodal Deep Learning with Clinical Metadata for Skin Lesion Classification." Scientific Reports 2026. (Accepted)
- **Wang, R.** "From Pro Leagues to Grassroots: R-VARS as the Future of Intelligent Football Refereeing." International Conference on Artificial Intelligence and Computational Intelligence. DOI: 10.1145/3730436.3730473. (Published)
- Su, C.*, Shen, Z.*, Guo, Y., **Wang, R.**, et al. "Arterial-Phase Liver MRI Synthesis via Dynamic Modality Adaptation and Multi-Scale Grouped Attention." IEEE Journal of Biomedical and Health Informatics. (Under Review)
- Shen, Z., **Wang, R.**, Su, C., Guo, Y., Liang, D., Shang, K. "CoSynSegNet: A Decoupled Collaborative Optimization Framework for Contrast-Enhanced MRI Synthesis and Tumor Segmentation." Journal of the Operations Research Society of China. (Under Review)
- Li, Z.*, Liu, F.*, Hong, Q., **Wang, R.**, et al. "OPERA: Offline Policy-guided Expert Routing and Adaptation for Universal Biomedical Image Analysis." ACM Multimedia (ACMMM) 2026. (Accepted)
- Liu, F., Saluja, R., Kwak, S., **Wang, R.**, Deng, R., Kim, H., Paetzold, J. C., Sabuncu, M. R. "MAdam: Metric-Aware Multi-Objective Adam." Conference on Neural Information Processing Systems (NeurIPS) 2026. (Accepted, Poster)

RESEARCH EXPERIENCES

Cornell University 12/2025 – Now

Visiting Student Researcher, SabLab, with Prof. Mert R. Sabuncu & Dr. Heejong Kim | Random Filter Loss

- Led RaFL (first author, ICLR 2027 submission): a random filter loss that compares prediction and target through random convolutions, random Fourier features and random wavelet filters, avoiding pretrained features and task-specific loss engineering.
- Balanced the filter families automatically via gradient-norm matching (no manual loss-weight tuning or grid search), and systematically studied filter family, weight-resampling schedule and initialization; the best configuration uses all three families, redraws filters every step and uses Gaussian initialization.
- Evaluated on three 2D and two 3D reconstruction tasks: outperforms a pixel loss on three and matches it on two, beats prior random-filter losses on all 2D tasks, and adds <2% to backbone forward FLOPs with no inference cost (182× cheaper than a VGG19 perceptual loss).

University of Washington 09/2025 – 12/2025

Project Collaboration, with Zihan Li | Foundation Models for Medical Imaging

- Fine-tuned multiple foundation-model backbones (EVA-X, ConvNeXt-V2, ViT, Swin-Transformer) using task-specific classification and segmentation heads, experimenting with LoRA, layer-freezing strategies.
- Built a systematic ensembling framework (logit-level averaging, weighted soft-voting, and stacking with a meta-learner) to combine heterogeneous models, achieving consistent gains in AUC/accuracy and robustness under distribution shift.

Research Center for Medical AI, SIAT, Chinese Academy of Sciences 11/2024 – 09/2026

Research Assistant, Prof. Kun Shang's Group | Medical Image Synthesis, Reconstruction & Segmentation

- *Arterial-phase liver MRI synthesis (JBHI, under review):* co-developed DMAMSA-Net, with a Dynamic Modality-Adaptive memory-selection system to resolve inter-phase ambiguity and a Multi-Scale Grouped Attention module (strip convolutions, linear complexity); +1–2 dB PSNR over ResViT/I2I-Mamba on LLD-MMRI/Duke with only 3.73M parameters.

- *Lightweight physics-guided MRI translation:* built a CNN that integrates analytical T1/T2 relaxation dynamics and predicts physical parameters instead of pixels (Sobel boundary constraints), removing the need for quantitative MRI priors; competitive fidelity on IXI/AIBL with 11.4M parameters (~10× smaller than SOTA, <8 GB VRAM).
- *Segmentation in complex scenarios (FSANet, first author, IEEE TIP under review):* proposed a Dual-Domain Awareness Module (FFT-based dynamic filtering) and an Edge Estimation Module; curated SceneX (11,000 images, 10 challenging scenarios); SOTA on 10 benchmarks, ~1.1% mIoU above SAM-2.
- *Other group projects:* 3D brain MRI synthesis (ProFA-GAN, PRCV 2026, co-first author), CT metal artifact reduction (SA-INR, IEEE SPL 2026, first author), and joint contrast-enhanced MRI synthesis & tumor segmentation (CoSynSegNet).

California Institute of Technology – SURF 2026 Fellow 06/2026 – 08/2026

Summer Undergraduate Research Fellowship, AnimaLab | 3D MRI Reconstruction with Neural Operators

- Selected for the 2026 Caltech Summer Undergraduate Research Fellowship (SURF); researched resolution-agnostic neural operators for 3D MRI reconstruction (related publication: NeurIPS 2026, co-first author).

University of California, Los Angeles 06/2024 – 09/2024

Research Student, Advisor: Prof. Xiang Anthony Chen | Intelligent Football Refereeing

- Proposed R-VARS, a refereeing framework integrating YOLOv8 and MViT for spatiotemporal foul classification, achieving 93.5% accuracy.
- Fine-tuned OpenLLaMA-8B via QLoRA to build an automated agent that generates match commentaries from visual event streams.
- Implemented multi-view fusion to aggregate features from diverse angles, resolving ambiguous offenses for grassroots league scenarios.

HONORS & COMPETITIONS

- **2026 Caltech Summer Undergraduate Research Fellowship (SURF), Caltech.**
- **IJCAI 2025 SpikeCV Infinity (Global 3rd Prize):** Integrated SNN with NeRF for sparse spike-stream reconstruction. Explicitly modeled in-scattering terms in the volume rendering equation to eliminate halo artifacts.
- **Electrical & Electronic Engineering Innovation Competition (Provincial 2nd Prize):** Designed "Daotian" smart grid system using LSTM for load prediction and dynamic algorithms for real-time power regulation.

WORK HISTORY

ByteDance 01/2026 – 03/2026

Computer Vision Algorithm Intern | Intelligent Vision AI: Face Recognition & Effects

- Built an end-to-end face recognition system using ArcFace (ResNet50) with 98.83% verified accuracy on LFW (6,000 pairs); implemented ONNX export and INT8 dynamic quantization achieving 4× model compression (174MB → 44MB) and 295 img/s GPU throughput.
- Trained StarGAN from scratch on 4K labeled faces for multi-attribute editing (gender/age swap) using WGAN-GP with gradient penalty, classification loss, and L1 reconstruction; additionally trained DCGAN for unconditional face generation with full training pipeline and loss curve analysis.
- Developed a real-time face effects pipeline (20–46 FPS on A6000) integrating landmark-driven AR stickers, bilateral-filter beauty filters with LAB-space whitening, style transfer (cartoon/warm/vintage), and 3DDFA_V2-based 3D face reconstruction (38K-vertex mesh with multi-angle rendering).

TECHNICAL PROFICIENCIES

- **Programming:** Python, Java, JavaScript/TypeScript, SQL, Bash, C++, Prolog, React
- **AI:** CV Low/High Level Tasks, Medical Imaging (MRI/CT), NLP (LLMs, QLoRA), Mamba/SSM, SNN.
- **Tools:** Git, Linux CLI, Docker, Postman, LaTeX, Medical Image Registration.
- **Certification:** iFLYTEK Prompt Engineering, DataWhale LeetCode Basic Algorithm, DataWhale Dynamic Programming Algorithm, Imperial College London (Coursera): Mathematics for Machine Learning.